



UNIVERSITY OF
CENTRAL FLORIDA

CAP 4053 - AI FOR GAME PROGRAMMING

Section: B001

College of Engineering and Computer Science

Department of Computer Science

Course Information

Term: Summer 2026

Class Meeting Days: TR

Class Meeting Time: 03:00PM - 06:50PM

Class Meeting Location: TCH 0103

Modality: P

Credit Hours: 3.00

Instructor Information

Johnathan Mell

Office Location: Discord

Office Hours:

After class or by appointment

Email: Johnathan.Mell@ucf.edu

Student Learning Outcomes

All outcomes apply to 5937 (graduate) students. Outcomes marked with a (G) only apply to graduate students, and the remainder make up the 4053 (undergraduate) list.

- **Learn how to make video games**
- **(G) Learn how to lead a video game design team**
- Identify and explain some areas of game programming that AI programmers handle that are not traditionally defined by academia as AI
- Explain what makes some Game AI “good” and other Game AI “bad”
- Use the Unity game engine and tutorials to create a simple game with AI, input and other interactive features
- Engage with your peers on Discord and communicate professionally about games
- Use basic vector operations to solve basic AI problems such as visibility cone computations
- Demonstrate understanding of the basic process of programming and how time and movement work in real-time systems
- **(G) Deploy multiple AI systems that serve different functions within your game**
- Identify and explain methods by which modern games implement decision making
- Demonstrate understanding of various pathfinding algorithms used in modern games
- Explain basic steering of entities, and how they achieve their desired effects
- Explain how game developers use focus testing to improve their AI **(G) and provide detailed analysis of that testing**

Course Description

CAP 4053 ENGR-COMP SCI 3(3,0)AI for Game Programming: PR: Complete all of the following Earn a minimum grade of C in each of the following: COP3502C - Computer Science I (3) or C.I. Surveys cutting-edge AI techniques for video games and board games and contrasts them with more traditional approaches. Spr

This is a High-Impact Course and will be noted as such on your transcript (undergrad only)!

- Classroom: <TBD>
- Course Modality: In-person*
 - We will use Discord extensively
 - We will use Zoom occasionally. See homepage for individual section links.
 - *see below regarding illness and recorded lectures
 - Past game streams are available on YouTube

This is a project-based course in which small groups will build a game, learning a great deal about how to do so in the process. This includes:

- learning quite a bit about programming in the practical sense
- learning both AI and “traditional” techniques for creating video games
- learning about what games ARE, and how to formally talk about them
- learning how to make games fun
- learning a small amount about the business and economics of game-making

Some groups will include a graduate student “team lead”. This team lead will be responsible for project managing the course, working with the other team members to deploy a more complex and ambitious project. This means as a graduate student, you will have additional responsibilities to manage your team, and the overall project will benefit from your involvement (although the expectations for the whole team will be commensurately higher).

The vast majority of your grade in this course is made up of several checkpoints in making a game. Doing a good job with making your game will mean you are almost certain to pass the course. Hopefully, you will also find the experience enjoyable!

Note: this class is exam-lite, but requires **a substantial amount of time** outside class. Especially if you are simultaneously enrolled in Senior Design, this can be daunting! Start early, and have regular meetings, with at least some in-person.

Required Course Materials and Resources

GameAI Pro

Notes: Optional resource! GameAI Pro is a reference guide, and generally I will not assign readings from it. It should be used to help guide your development, and reading it is recommended to help elaborate on lecture content.

Online Access: <https://www.gameaipro.com/>

Class Schedule

See the Webcourses homepage for the most up to date schedule.

Course Assessment and Grading Procedure

I use an “only-good curve” in my class. That means that I may choose to curve the class in your favor (e.g., by shifting the requirement to get an A to a lower percent). But I will not shift the curve away from your favor. This means that if every student gets 94% or above, then you will all get ‘A’s (this would be my preference!).

We always evaluate games and presentations with multiple viewers and instructors to reduce the impact of opinion or personal preference. **That said, a good game that does not otherwise stand out is expected to be around a "B".** "A" games are outstanding products, and "C" games are functional but flawed in several ways. "D" games are deeply problematic or lacking features, and "F" and below games are nearly unplayable or truly trivial efforts.

This is a group project. Some of your grade comes from individual assessment, but the majority comes from team assessments. It is highly unlikely that all members of your team will contribute equally, but peer assessments are included in order to mitigate this effect. That said, the intent is for you to succeed or fail as a team, as in real-world projects.

You will note that a substantial part of your grade is peer review. This means that you will be required to rate your partners by the amount of effort, skill, and help that they brought to creating the game project. This will be required just after each of the three main game project deliverables. You must do your peer reviews at this time, but the peer reviews grades will not be visible until the end of the class (to avoid drama!). Actual text of peer reviews will be released on request.

Undergraduate grade breakdown:

Assignment	Sub-part	Percent of Final Grade
Game Project		62%
	Game Design Document	15%
	Mid-semester presentation (playable prototype)	15%
	Final presentation (full game) + executable binary: score comes from BOTH parts	30%

	Status Report	2%
Quizzes		6%
	Quiz #1	3%
	Quiz #2	3%
Assignments		12%
	Board Game Showcase	3%
	Unity Practicum	3%
	Testing Assignment	3%
	Level Design Assignment	3%
Peer Review, Final Attendance	(Final peer review is worth double)	20%
Total		100%

Graduate grade breakdown:

Assignment	Sub-part	Percent of Final Grade
Game Project		60%
	Game Design Document	10%
	Mid-semester presentation (playable prototype)	10%
	Final presentation (full game) + executable binary: score comes from BOTH parts	30%

	Biweekly status reports	10%
Quizzes		6%
	Quiz #1	3%
	Quiz #2	3%
Assignments		14%
	Board Game Showcase	2%
	Unity Practicum	2%
	Testing Assignment	2%
	Level Design Assignment	2%
	Technical Documentation Assignment	3%
	Analysis of Playtesting Results	3%
Peer Review, Final Attendance	(Final peer review is worth double)	20%
Total		100%

Grading Scale

Letter Grade	Percent range
A	94-100
A-	90-93
B+	87-89
B	84-86
B-	80-83
C+	77-79
C	74-76
C-	70-73
D+	67-69
D	64-66
D-	60-63
F	<60

Past Grades

In the past, my grade distribution has been:

A	30.17%
A-	29.38%
B+	17.38%
B	11.69%
B-	8.85%
C+	0.95%
C	1.11%
C-	0.16%
D+ or less	0.32%

Communication Expectations

Message me on Discord or catch me in class. I recommend using @Instructor on Discord rather than @Prof. Mell for a faster response. Responses are normally within minutes to hours, but please allow up to a day.

Policies for Course Grade

Makeup Work Policy/Missed/Late Assignments

Per university policy, you are allowed to submit make-up work (or an equivalent, alternate assignment) for authorized university-sponsored activities, religious observances, or legal obligations (such as jury duty). If this participation conflicts with your course assignments, I will offer a reasonable opportunity for you to complete missed assignments. The make-up assignment and grading scale will be equivalent to the missed assignment and its grading scale. In the case of an authorized university activity, it is your responsibility to show me a signed copy of the Program Verification Form for which you will be absent, **prior** to the class in which the absence occurs. In any of these cases, please contact me **ahead of time** to notify me of upcoming needs. **Missing without advanced notice** is handled on a case-by-case basis, but may result in zero credit!

Attendance

I will not generally be taking attendance as part of your grade, **except** the assignment work days, during which part of your grade counts toward assignments. Additionally, there will be required quizzes on the schedule, as indicated below. These quizzes may be taken remotely any time during a 24-hour period, and cannot be made up after the fact. There will be at least one 1-hour block during which TAs/graders will be available over Discord for assistance and questions. Reasonable consideration will be given to circumstances of illness or personal emergency, when stated **before** the quiz. Forgetfulness is not a personal emergency.

Some lectures featuring guest lecturers will be conducted over Zoom. We also have a Zoom link for situations when I cannot be in class (e.g. due to illness or travel). While I will make the slide decks available after class, I will generally **NOT make recordings** available of the lectures. I do not "read from the slides" so this means if you don't attend class, you will miss a good section of the course. I make no guarantees that all quiz material will be on the slides themselves; I will make assurances that I will cover such material in class.

Make-up Assignments for Authorized University Event or Co-curricular Activities

Students who represent the university at an authorized event or activity (for example, student athletes) and who are unable to meet a course deadline due to a conflict with that event must provide the instructor with documentation in advance to arrange a make-up. No penalty will be applied if the student gives advance notice and communicates with the instructor following UCF policy. In the case of an authorized university activity, it is your responsibility to show the instructor a signed copy of the Program Verification Form for which you will be absent, prior to the class in which the absence occurs or due date you need extended. For more information, see [UCF Policy 4-401](https://ucf.edu/policy/4-401).

Make-up Assignments for Religious Observances

A student who desires to observe a religious holy day of his or her religious faith must notify their instructor as soon as practicable prior to the observance. Students who are absent because of religious observances will be permitted a reasonable amount of time to complete any missed work. For more information, see [UCF Regulation 5-020](#).

Artificial Intelligence (AI) Use Policy

Use of Generative AI (GenAI) is freely permitted but must be disclosed. Students are allowed to use Generative Artificial Intelligence (GenAI) tools on assignments if the usage is properly disclosed and credited. For example, text generated from Copilot should include a citation such as: "Copilot. Accessed 2023-12-03. Prompt: 'Summarize the Geneva Convention in 50 words.' Generated using Copilot." Any attempt to represent GenAI output inappropriately as your own work will be treated as plagiarism. Ultimately, students are responsible for the accuracy and veracity of submitted work whether AI-generated or not.

AI, LLMs, and GPT-X

I love AI. This is an AI course. If you choose to use stable diffusion, large language models, or other AI techniques to create voice clones, write adaptive narrative or dialogue, or create art for your game, please do so! I ask that you merely cite that you have done so. Do not pass off such work as your own; you will miss out on the credit I will give you for having utilized modern tools for modern solutions.

Copyright and IP

Professor's Copyright:

This course may contain copyright protected materials such as audio or video clips, images, text materials, etc. These items are being used with regard to the Fair Use doctrine in order to enhance the learning environment. Please do not copy, duplicate, download or distribute these items. The use of these materials is strictly reserved for this online classroom environment and your use only. All copyright materials are credited to the copyright holder, either me, Dr. Mell, or the other relevant holder.

Students' Copyright:

Your games that you create are your own copyright. You should consult with a legal expert to fully explain what this means, but generally, it means you can use your games in portfolios and other items, including commercial use. If you use 3rd-party materials in your game (e.g., music or art), you are still subject to those terms, and the terms of any game engine you may use. I reserve the right to use clips of your games in future classes for educational and promotional purposes. I additionally reserve the right to make available a copy of your game binary (executable) on my website for archival purposes and to

demonstrate the class. Source code, while you will be required to submit it, will not be made publicly available. If you would prefer your final game to not be hosted publicly, or you'd instead like to provide a live link to your own site (e.g., Itch.io), please contact me directly.

Illness and Class

COVID/Illness Special Note from Dr. Mell

COVID-19 has changed the way public health is handled worldwide, indefinitely.

UCF's official policy has been that you are expected to be up to date and boosted with COVID-19 vaccine if you are able, and to otherwise follow CDC guidance.

To be considerate of the health and time of others, I do not offer regular in-person office hours. Largely, this has not been a problem--I am normally available on Discord very readily with fast response times. However, if you believe you need an exception, please reach out to me and we can discuss.

I am trying to make this class as interactive as possible while still respecting student's individual needs. That is why I have chosen not require attendance beyond the major points of the class. I hope you choose to safely engage as much as possible in spite of this, and I will work with you to find good options with regard to your health and safety.

If you are ill, do not come to class. If you have a presentation, please notify me before class and we will offer alternatives as appropriate.

Recording

Dr. Mell's summary: don't give away recordings without my permission

Students may, without prior notice, record video or audio of a class lecture for a class in which the student is enrolled for their own personal educational use. A class lecture is defined as a formal or methodical oral presentation as part of a university course intended to present information or teach students about a particular subject. Recording classroom activities other than class lectures, including but not limited to lab sessions, student presentations (whether individually or part of a group), class discussion (except when incidental to and incorporated within a class lecture), clinical presentations such as patient history, academic exercises involving student participation, test or examination administrations, field trips, and private conversations is prohibited. Recordings may not be used as a substitute for class participation and class attendance, and may not be published or shared without the written consent of the faculty member. Failure to adhere to these requirements may constitute a violation of the University's Student Code of Conduct as described in the Golden Rule .

Third-Party Tools

Unity Game Engine Privacy Statement

You may install Unity, and have access to the pro version, by simply signing up on the Unity website with your UCF.edu email address (you will receive a student version).

Students have also used Unreal Engine and Godot Engine in the past, successfully. However, I don't officially "support" them for this course.

Third-Party Software and FERPA

During this course you might have the opportunity to use public online services and/or software applications sometimes called third-party software such as a blog or wiki. While some of these could be required assignments, you need not make any personally identifying information on a public site. Do not post or provide any private information about yourself or your classmates. Where appropriate you may use a pseudonym or nickname. Some written assignments posted publicly may require personal reflection/comments, but the assignments will not require you to disclose any personally identity-sensitive information. If you have any concerns about this, please contact your instructor.

Disability Access & Accommodations

The University of Central Florida is committed to providing equal access to all students with disabilities (ADHD, learning disabilities, Autism, chronic medical conditions, physical disabilities, etc.). To receive consideration for reasonable disability-related course accommodations, disabled students must contact Student Accessibility Services (SAS) and complete the steps required for SAS to review accommodation requests. More information can be found on the UCF [Student Accessibility Services](#) website under the Start Here tab or by contacting SAS directly (Ferrell Commons 185; sas@ucf.edu; Phone - 407-823-2371).

Approved accommodations are shared with course instructors via the SAS Course Accessibility Letter. Implementing certain accommodations may require discussion and specific considerations of the course design, course learning objectives, and the individual academic and course challenges experienced by the student. While students with disabilities or chronic health needs are also encouraged to discuss any course concerns with professors in addition to contacting SAS, professors are not required to facilitate disability-related adjustments to the course unless the professor has received a Course Accessibility Letter from SAS that outlines approved accommodations.

Notice

Note: Everything below this point is UCF policy, and was not specified by the instructor. Please refer to my specific academic integrity notes above, which override some of the boilerplate.

Academic Integrity

Students should familiarize themselves with UCF's Code of Conduct at [Student Conduct and Integrity Office](#). According to Section 1, "Academic Misconduct," students are prohibited from engaging in:

- a. Academic misconduct is defined as any submitted work or behavior that obstructs the instructor of record's ability to accurately assess the student's understanding or completion of the course materials or degree requirements (e.g., assignment, quiz, and/or exam). Examples of academic misconduct include but are not limited to: plagiarism, unauthorized assistance to complete an academic exercise; unauthorized communication with others during an examination, course assignment, or project; falsifying or misrepresenting academic work; providing misleading

information to create a personal advantage to complete course/degree requirements; or multiple submission(s) of academic work without permission of the instructor of record.

- b. Any student who knowingly helps another violate academic behavior standards is also in violation of the standards.
- c. Commercial Use of Academic Material. Selling of course material to another person and/or uploading course material to a third-party vendor without authorization or without the express written permission of the University and the instructor of record. Course materials include but are not limited to class notes, the instructor of record's slide deck, tests, quizzes, labs, instruction sheets, homework, study guides, and handouts.
- d. Soliciting assistance with academic coursework and/or degree requirements. The solicitation of assistance with an assignment, lab, quiz, test, paper, etc., without authorization of the instructor of record or designee is prohibited. This includes but is not limited to asking for answers to a quiz, trading answers, or offering to pay another to complete an assignment. It is considered Academic Misconduct to solicit assistance with academic coursework and/or degree requirements, even if the solicitation did not yield actual assistance (for example, if there was no response to the solicitation).

Responses to Academic Dishonesty, Plagiarism, or Cheating

Students should also familiarize themselves with the procedures for academic misconduct in UCF's student handbook, [The Golden Rule](#). UCF faculty members have a responsibility for students' education and the value of a UCF degree, and so seek to prevent unethical behavior and respond to academic misconduct when necessary. Penalties for violating rules, policies, and instructions within this course can range from a zero on the exercise to an "F" letter grade in the course. In addition, an Academic Misconduct report could be filed with the Office of Student Conduct and Academic Integrity, which could lead to disciplinary warning, disciplinary probation, or deferred suspension or separation from the University through suspension, dismissal, or expulsion with the addition of a "Z" designation on one's transcript.

Being found in violation of academic conduct standards could result in a student having to disclose such behavior on a graduate school application, being removed from a leadership position within a student organization, the recipient of scholarships, participation in University activities such as study abroad, internships, etc.

Title IX

Title IX prohibits sex discrimination, including sexual misconduct, sexual violence, sexual harassment, and retaliation. If you or someone you know has been harassed or assaulted, you can find resources available to support the victim, including confidential resources and information concerning reporting options at [Let's Be Clear](#) and [UCF Cares](#).

For more information on access and community engagement, Title IX, accessibility, or UCF's complaint processes contact:

- Title IX – ONAC – [Office of Nondiscrimination & Accommodations Compliance](#) & askanadvocate@ucf.edu
- Disability Accommodation – Student Accessibility Services – [Student Accessibility Services](#) & sas@ucf.edu

- [Access and Community Engagement](#) (including the Ginsberg Center for Inclusion and Community Engagement, Military and Veteran Student Success, and HSI Initiatives)
- UCF Compliance and Ethics Office – [Compliance, Ethics, and Risk Office](#) & complianceandethics@ucf.edu
- The [Ombuds Office](#) is a safe place to discuss concerns.

Reporting an Incident or Issue

If you believe you have experienced discrimination by any faculty or staff member, contact the Office of Nondiscrimination & Accommodations Compliance via the [ONAC website](#) or at 407-823-1336. You can also choose to report using the UCF Integrity Line either anonymously or as yourself at 1-855-877-6049 or by using the [online form](#). UCF cares about you and takes every report seriously. For more information see the [Reporting an Incident or Issue Webpage](#).

Deployed Active-Duty Military Students

Students who are deployed active duty military and/or National Guard personnel and require accommodation should contact their instructors as soon as possible after the semester begins and/or after they receive notification of deployment to make related arrangements.

Campus Safety

At UCF Public Safety and Police, safety is the top priority. Emergencies on campus are rare, but if one should arise, it's important to be familiar with some basic safety and security concepts.

- In an emergency, always dial 911.
- Every UCF classroom has an **Emergency Procedure Guide** posted on a wall near the door, which will show you how to respond to a variety of situations. This guide can also be found online [here](#).
- In the event of an active threat, remember **AVOID, DENY, DEFEND**. Choose the best course of action and act immediately. Watch the video [here](#) to learn more.
 - **AVOID.** Pay attention to your surroundings and have an exit plan. Get as much distance and as many barriers between you and the threat as quickly as possible.
 - **DENY.** When avoiding is difficult or impossible, deny the threat access to you and your space. Lockdown by creating barriers, turning the lights off and remaining quiet and out of sight. Make sure your cell phone is silenced, but do not turn it off.
 - **DEFEND.** When you are unable to put distance between yourself and the threat, be prepared to protect yourself. Commit to your actions, be aggressive and do not fight fairly. Do whatever it takes to survive.
- For emergencies on campus, UCF will utilize the [UCF Alert](#) system. All UCF students, faculty and staff are automatically enrolled to receive these email and text alerts, however, it's a good idea to frequently ensure your [contact information is up to date](#).

Financial Aid Accountability

All instructors/faculty are required to document students' academic activity at the beginning of each course. In order to document that you began this course, please complete this activity by the end of the first week of classes or as soon as possible after adding the course. Failure to do so may result in a delay in the disbursement of your financial aid.